openSUSE Asia Summit 2026
Yogyakarta, October 3 - 4

# State of SUSE Linux Enterprise: What's new in 16.1

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# Who are we




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# Agenda

## Why?

- Why commercial Linux complements free open source, rather than betraying it

## What?

- What's in SLES 16: core technologies, security and developer tools

## How?

- Built in the open: OBS, openQA and the supply chain

## The lifecycle

- Leap and SLES lifecycles: from 24 months to 16 years

## What’s new in 16.1 & 16.2

- Immutable mode, real time in the OS, kernel and tools, open agentic AI

# Why?

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# Commercial and Free: Not a Contradiction

*Commercial Linux: not the enemy of free software, but one of its engines*

1

### The code stays free

SLES sources are public, and the GPL itself allows charging for distribution and support.

2

### We sell what code cannot give

Support, certifications, a 16-year lifecycle and someone accountable when it breaks.

3

### The revenue flows back

It pays engineers working upstream, runs OBS and openQA, and Leap shares the SLES binary core.

Those who need guarantees pay for them, and fund the work that everyone gets for free

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# From Community Code to Mission Critical

*Enterprise Linux is how open source reaches systems that must not fail*

**90%+**

of the world's leading companies rely on SUSE

**10,000+** enterprise customers

**14/15**
largest aerospace companies

**13/15**
largest financial services firms

**10/10**
largest automotive makers

**13/15**
largest pharmaceutical companies

**Certified and supported:** that is how your commit reaches banks, power grids and aircraft.

*Source: SUSE company data*

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# Digital Public Good

*The first and only commercial Linux recognized by the Digital Public Goods Alliance*

- Independently vetted against the DPG Standard: open licences, privacy, do no harm
- Open, transparent, community-aligned digital infrastructure
- Strengthens digital sovereignty, trust and global collaboration

**Proof that commercial and open can be the same thing.**

Digital Public Goods Alliance

DPG

# What?

# Built on Industry Standards

*The building blocks admins already know from any Linux*

- Select Core Components
  - Upstream kernel with backports and security patches
  - Upstream Base Components (systemd, glibc etc.)
  - Upstream Security Components (OpenSSL, OpenSSH etc.)
- Adopt Mainstream technologies
  - Cockpit web console and Ansible for management
  - SELinux as the default security framework
  - NetworkManager for networking
- Reducing skill gap: easy for admins: we use the popular upstream technologies

# Plus SUSE-Driven Open Source

*Projects SUSE leads or champions, added where they make sense*

- **Btrfs:** copy-on-write filesystem with OS-level snapshots, any footprint
- **Snapper:** simple snapshot management, see what changed and roll back
- **Agama:** web-based, API-driven installer, easy to fully automate
- **Kernel Live Patching:** kernel fixes without a reboot (kpatch + kGraft)
- **User Space Live Patching:** patch selected libraries without restarting services

# Certifications & Compliance

*What each one means, and who is behind it*

| Program | What is it about? | What does it mean? | Body |
|---|---|---|---|
| **Common Criteria** | “How can I be sure to get the security I need?” | · Whole build system, including:<br>· Installation & Processes<br>· Build IT environment<br>· Physical environment | **BSI**<br>**NIAP** |
| **FIPS 140-3** | “Are my crypto algorithms conforming to the standard?” | · Crypto Libraries<br>· Random number generators | **NIST** |
| **STIG** | “I need a document formalizing a hardening guide!” | · Needs to look at the whole system following the questions / guidelines of the DISA | **DISA** |
| **PCI DSS** | “I need to manage financial data (e.g. credit cards)!” | · Affects more than the OS or software: hardware, datacenter, ... | **SSC** |

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# Protecting Data in Every State

*Confidential computing closes the last gap: data in use*

## Data at rest

- Storage encryption
- Disks and volumes stay unreadable if stolen

## Data in transit

- Network encryption
- Identity-based access with least privilege

**Confidential computing**

## Data in use

- Memory and CPU state encrypted
- Remote attestation proves the environment before data is released

**Zero trust across all three:** verify the user, the network and the hardware (remote attestation)

**For openSUSE:** Leap 16 inherits it through the shared core, and COCONUT-SVSM, started by SUSE and now a Linux Foundation project, is open to contributors.

# Catering to Developers!

*Packaging and maintaining the needed tools for them …*

## Java · *Java on Java!*

- Java and Apache Tomcat included

## Programming Languages

- Python, Go, Ruby, Perl, C, C++, PHP … updated

## Other components

- PostgreSQL, MariaDB are included

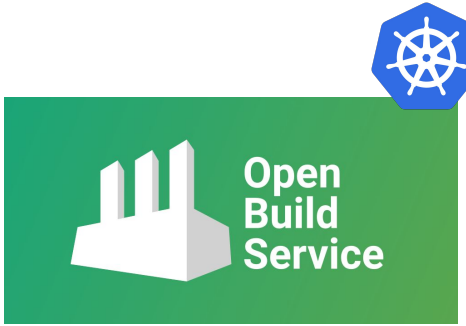
# How?

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# Open Builds Service

**Trust: fully open source, for SUSE and the community**

- Builds SLES, openSUSE and the Kubernetes packages
- Full visibility into how SUSE builds its products
- Used by the community too: Tumbleweed, almost daily
- "Toolified" process: certifications build on it




**Supply chain security built in:** *from source to product, with proof*

**Sources**
Git repos, tarballs, upstream projects

**OBS: build & test**
Stores sources, binaries and signing keys; dependency tracking; reproducible builds

**Products**
Packages, ISOs / VMs, containers

**Proof**
SLSA provenance, SPDX and CycloneDX SBOMs

# Open Testing

*openQA*

Open Source makes projects innovate and mature faster.

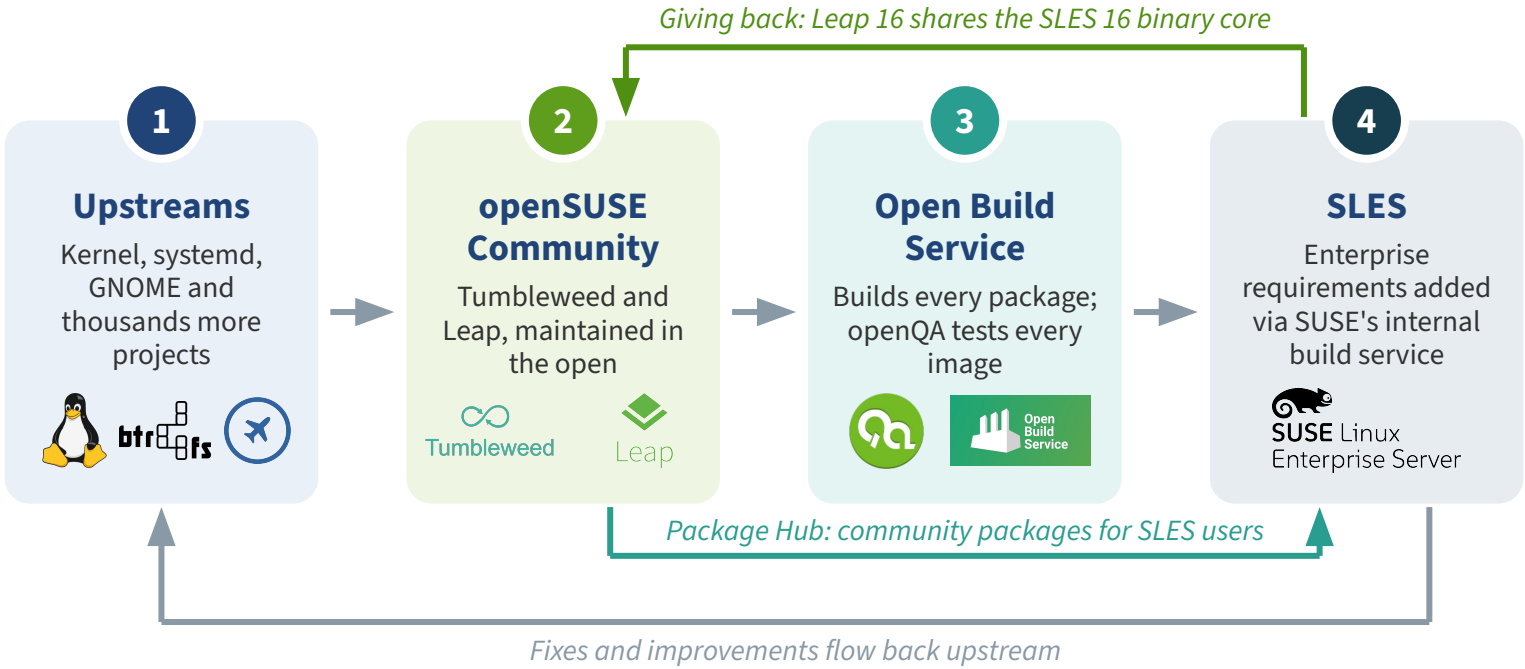
Our Quality Assurance and Quality Engineering are open source.



## openQA

- Open Auditable Quality Assurance
- Ensures interoperability across versions
  - Eases Minor and Major version upgrades
- Enables a predictable yearly release cadence

# SUSE Linux is True Open Source



# Our Linux Menu ...

SUSE Linux Enterprise Server 16

- Enterprise oriented
- Certified
- Supported
- Long Life Cycle

openSUSE Leap

- Community oriented
- Perfect to prototype, learn or self-support
- Convertible to SUSE Linux Enterprise Server

openSUSE Tumbleweed

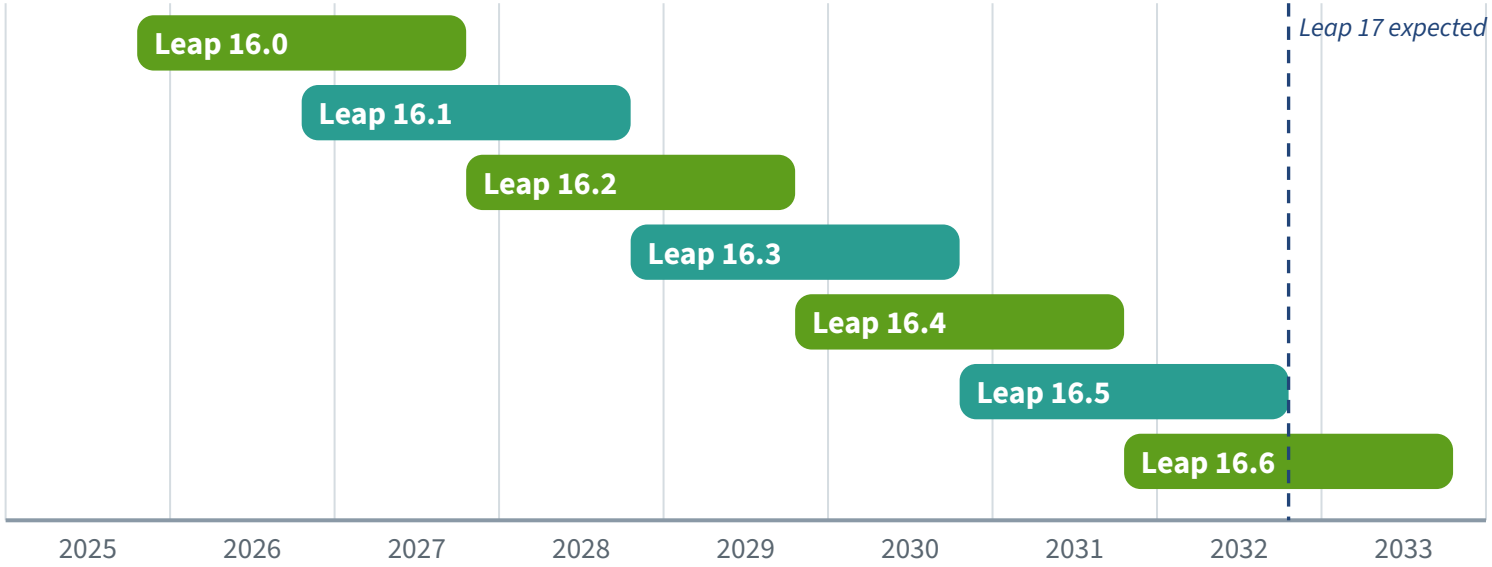
- Community oriented
- Rolling-Release
- Always up to date
- Perfect to experiment

# Lifecycle

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# Leap 16 lifecycle

*One release a year, each supported for 24 months*

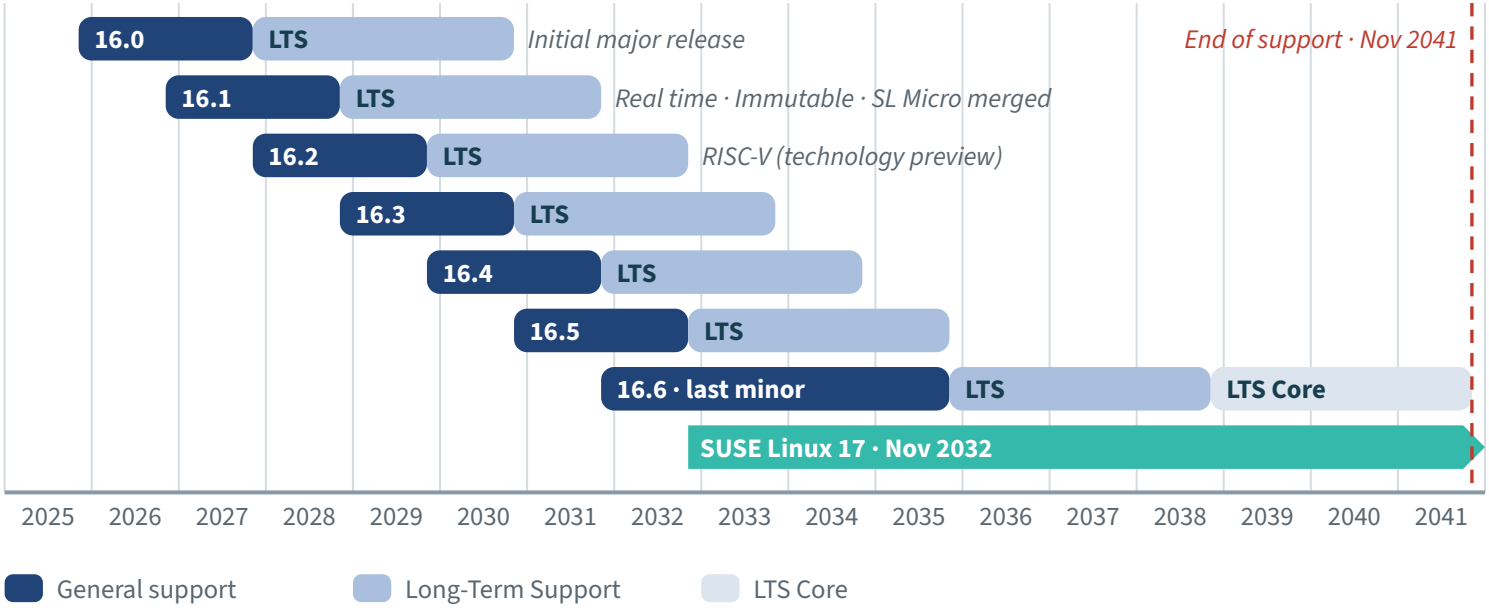


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# Need to stay longer?

Leap 16.1 → SLES 16.1 migration gives you that chance.

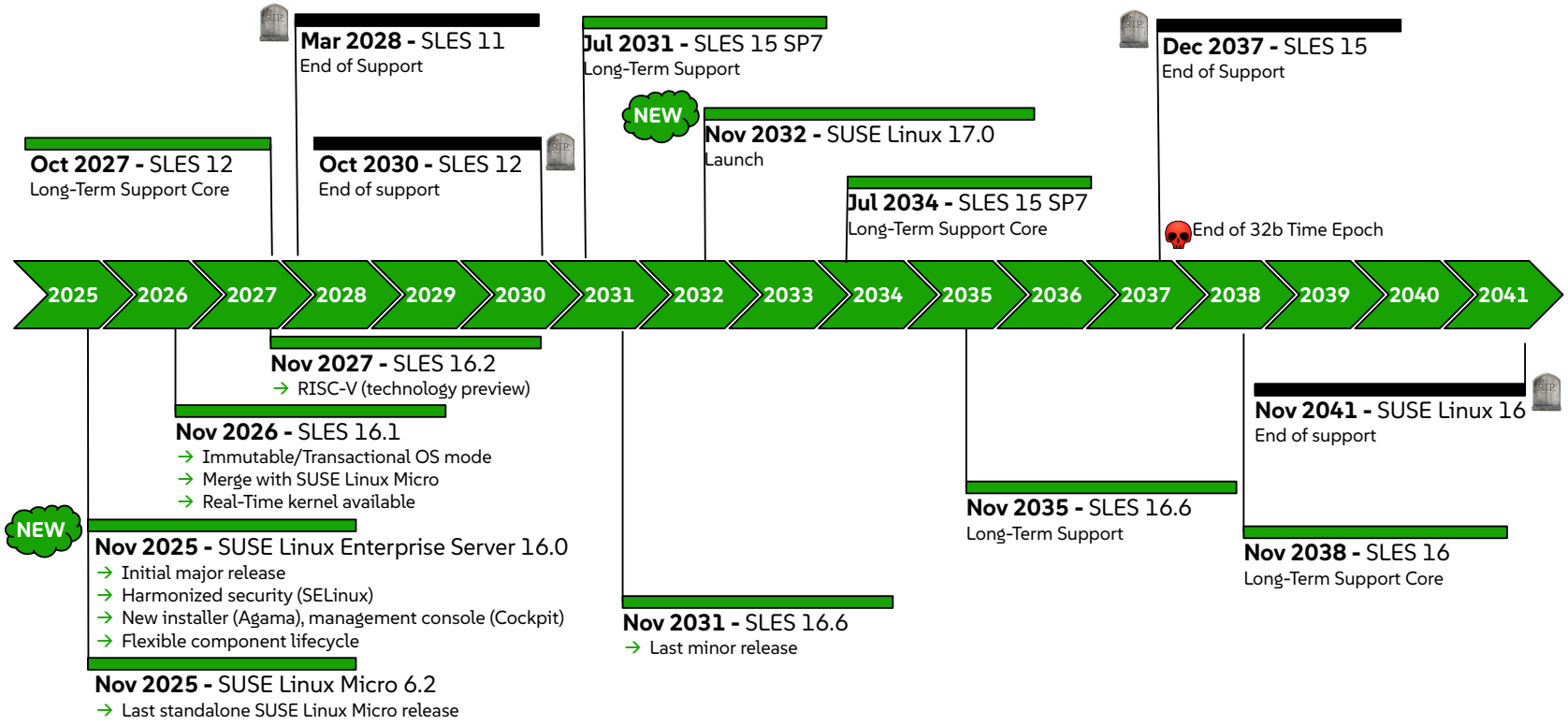
Move over any time with the opensuse-migration-tool.



# SUSE SLES milestones

*“Memento mori” - all the good things come to the end at some point of time*

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# Flexible component lifecycles in 16




Different approaches to lifecycle for individual OS components, to meet a variety of needs for stability and feature velocity.

## Stable

* Components that remain stable and not change during the entire SLES 16 lifecycle
* Change is minimal and focused on stability over new features
* Updates being fully backward compatible

*Examples: glibc, util-linux, perl*

## Balanced

* Components that are updated to align with upstream evolution and customer demand
* Smooth transition path, there's a way to get the original behavior back or there's the old version in parallel for some time

*Examples: systemd, kernel, mariadb, postgresql, hardware enablement*

## Agile

* Components that are under active development
* Packages that are reasonably close to the current upstream versions
* E.g. a "sliding window" of releases, offering both the current and a recent older version

*Examples: go, rust, gcc, public cloud tools, timezone*

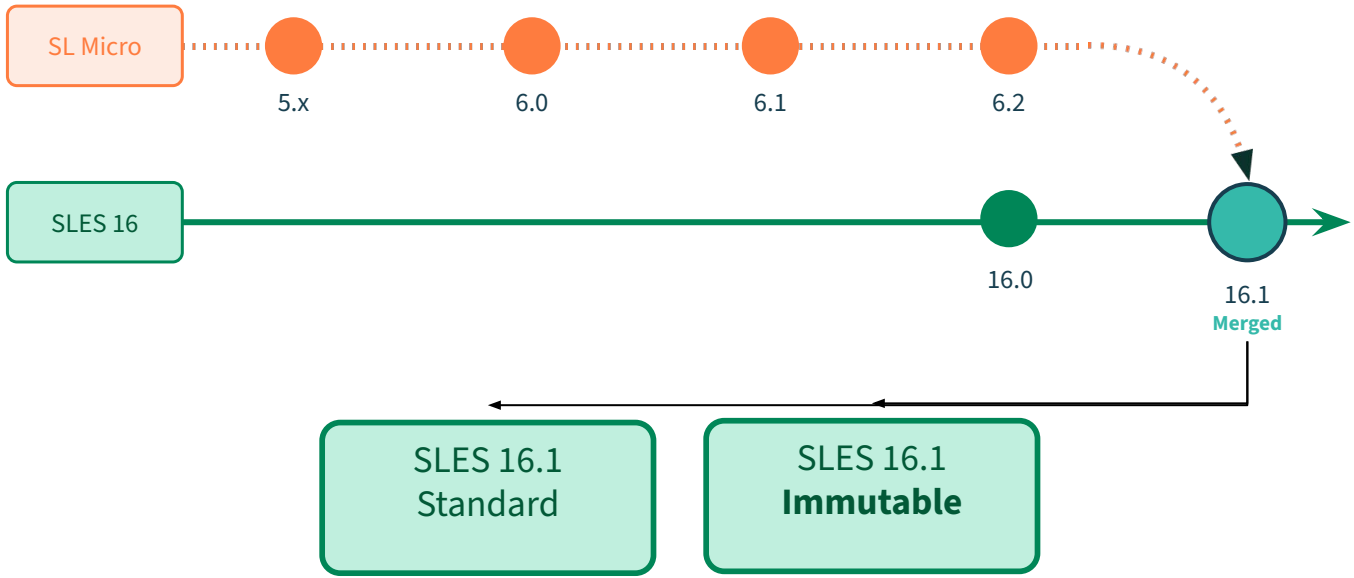
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# What’s new in 16.1

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# Immutable Mode for SLES

*Merging SL Micro into SLES*



# SLES Is Now a Real-Time OS

*The PREEMPT_RT kernel moves from a separate product into SLES itself*

## What it is

- Predictable response times for time-critical workloads
- Improves the worst-case latency, not overall throughput
- Warning: Soft real time only!  hard real time is not for SUSE (or any other Linux)

## Why I should care

- PREEMPT_RT landed in mainline Linux 6.12, the kernel SLES 16 is built on
- A solid driver program makes real-time driver work easier

**1 Install SLES 16.1**
with the default kernel

↓

**2 Switch to the RT kernel**
whenever you need it

↓

**3 No extra subscription**
included with SLES

*Migration: SLE RT 15 SP7 → SLES 16.1*

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# Kernel 6.12, Refreshed

*Upstream features developers and tinkerers will notice*

## Performance

- PREEMPT_LAZY replaces PREEMPT_VOLUNTARY: better efficiency for high-throughput workloads
- libfabric 2.5: faster RDMA with less CPU overhead

## Build and debug

- eBPF CO-RE out of the box: vmlinux.h packaged, no custom toolchains
- ublk: block devices in user space via io_uring
- pcp 7.9: richer telemetry for root-cause analysis

## Trust and lifecycle

- Better TPM handling for Secure Boot and disk encryption
- fwupd-efi: safer, automated firmware updates
- NVIDIA open GPU kernel drivers: fewer crashed AI jobs

**Plus new hardware:** Arm (AmpereOne, AGI CPU), AMD Zen 6, NVIDIA Vera

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# Systems Management

*Agama, Cockpit and Ansible in 16.1*

## Agama installer

- Standard or Immutable, chosen at install time
- Self-update: pulls the latest fixes at boot
- Day-0 driver updates for brand-new hardware
- UI for bonding, bridges, VLANs and LVM

## Cockpit web console

- Snapper snapshots: view, compare and boot from them
- SELinux policy UI plus setroubleshoot for denials
- Transactional updates for immutable systems

## Ansible automation

- Ansible Core 2.20: stricter, clearer failures
- New system roles: kernel settings, sudo, sshd, postgres, network, logging
- Dev tools: ansible-lint, molecule and more

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# Spice it up: Open Agentic AI for IT Ops

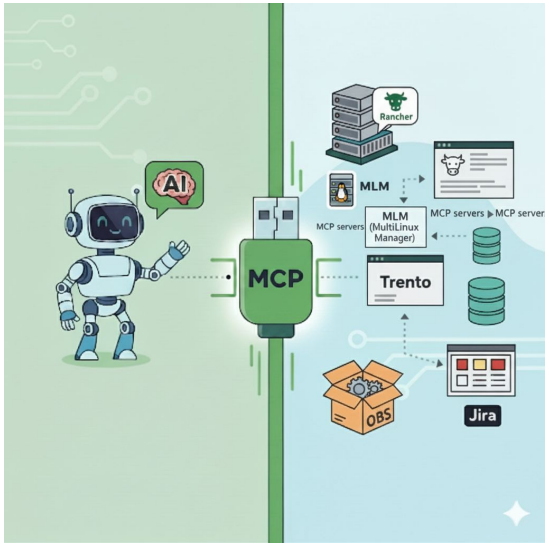
*Use your AI of choice to improve operations in an open standards way*

## AI-ready Linux:

- Embedding AI capabilities using Model Context Protocol standard

## How?

- Agentic AI = AI that acts
- MCP = the plug that enables it
- Together → smarter workflows



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# SLES 16.2: what is over the corner?

# What's coming in 16.2

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01

## Immutable Mode Improvement

***Immutable without compromise***

Sell immutable as the mainstream default: same admin experience, 30-second rollback, zero config drift across fleets.

02

## Image Mode

***Manage the OS like your containers***

Sovereign, air-gapped OS lifecycle in the customer's own CI/CD — the differentiator vs. RHEL bootc.

03

## AI

***Supervised by default, autonomous on your terms***

On-box, auditable AI that bridges the admin skills gap — not a SaaS report. Clear win vs. RHEL Insights.

04

## Security & Compliance

***Trustworthy by construction***

Shipped — not roadmapped. Clears procurement gates for NIS-2, CRA, US Federal and BFSI buyers.

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# Q&A

# More

*To learn more about SLES 16*

**SLES 16 Official Admin Book:**

- https://amzn.to/3ZeDtn6

**But would be better just to try SLES 16:**

- https://www.suse.com/download/sles/

# Thank You



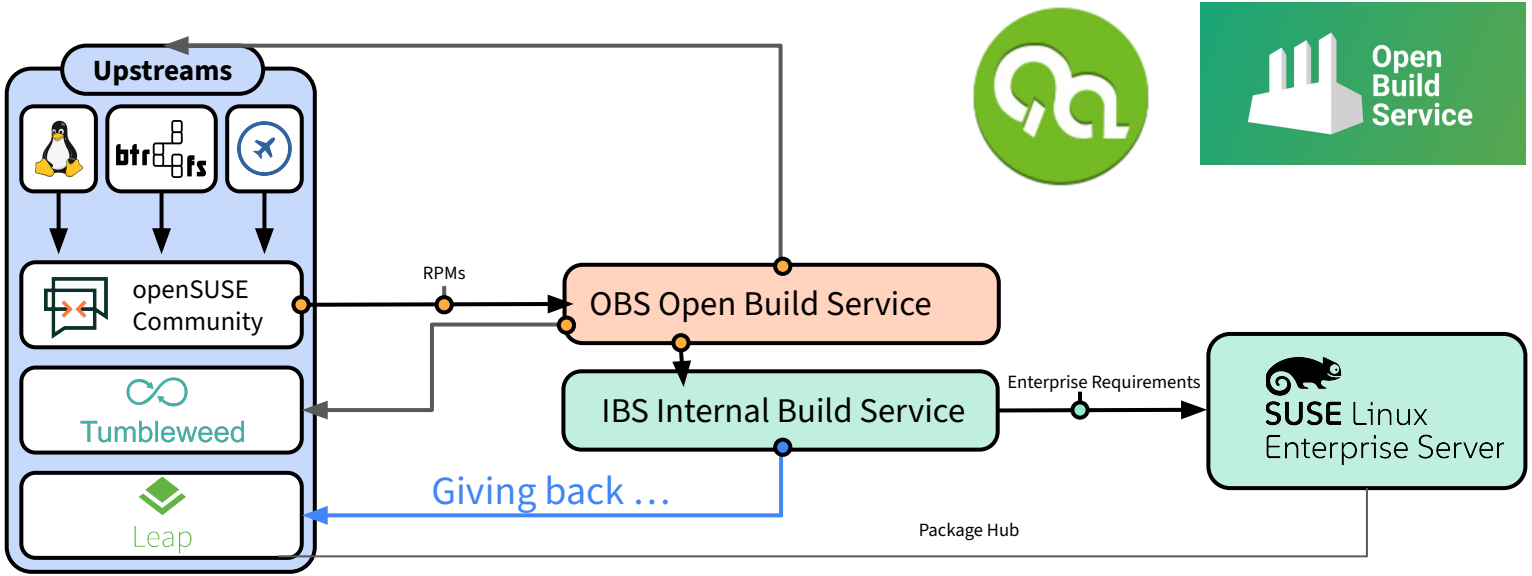
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# Appendix - support slides

# Distro dependencies

*The network of distro dependencies - a flow chart diagram*



# The Value of SUSE Subscriptions and Support

SUSE goes beyond successful implementations of open source

